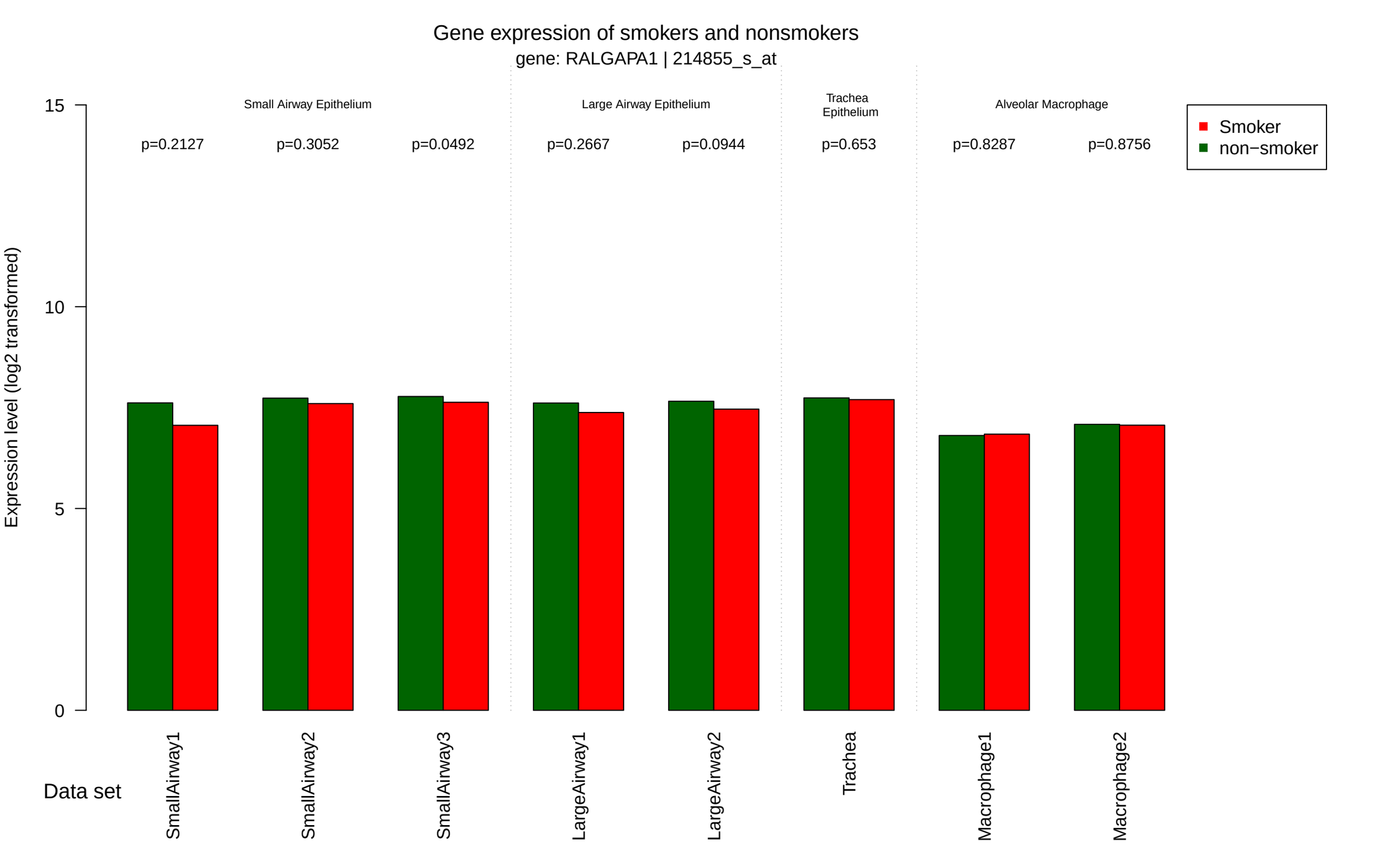
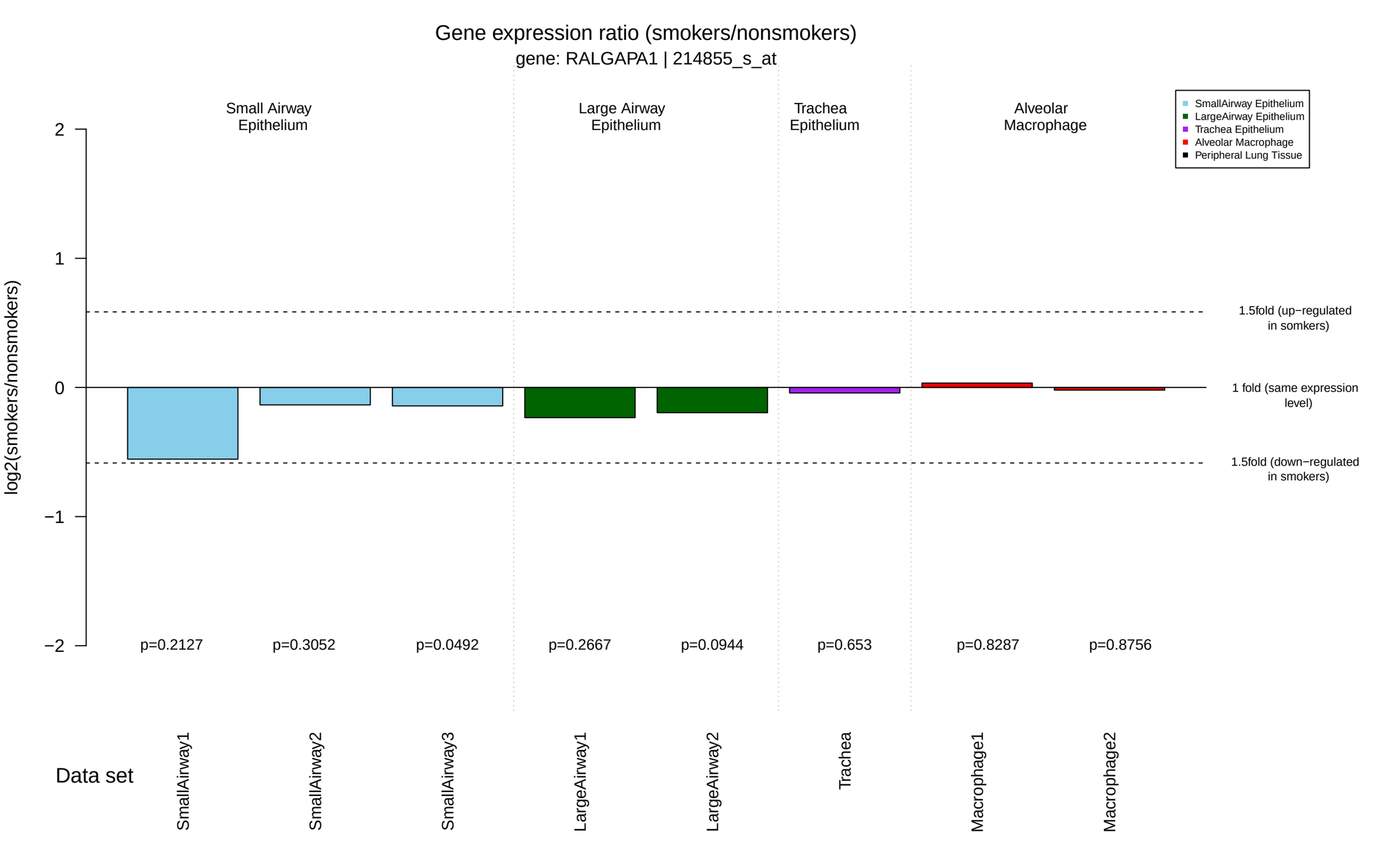
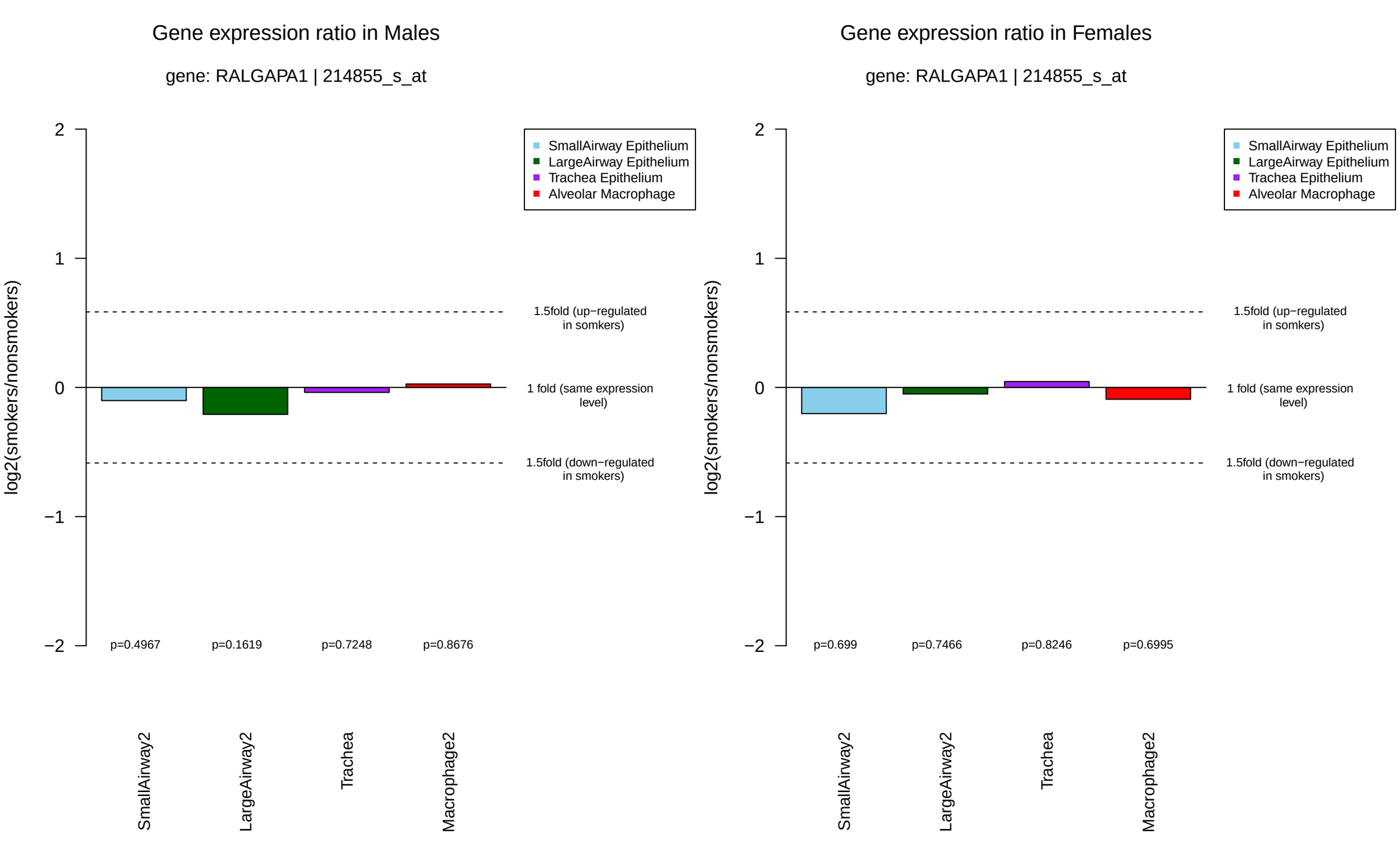
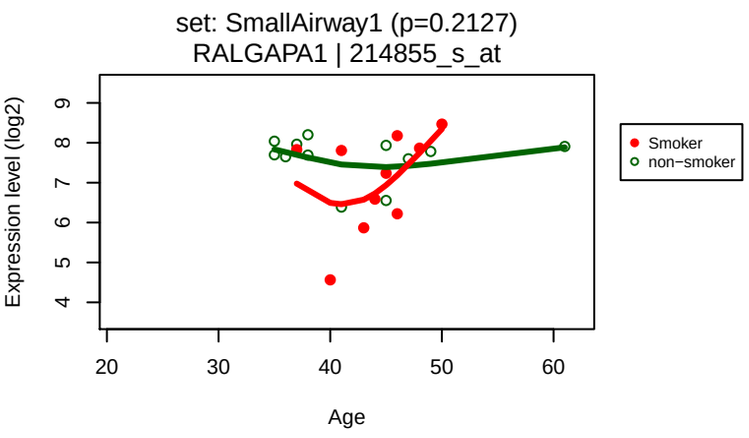
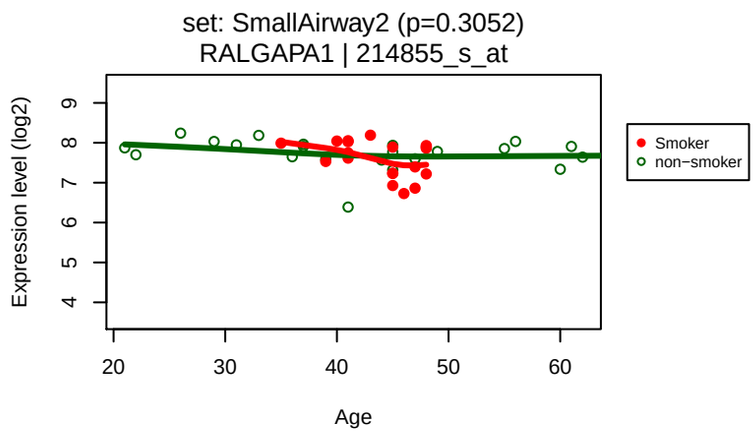
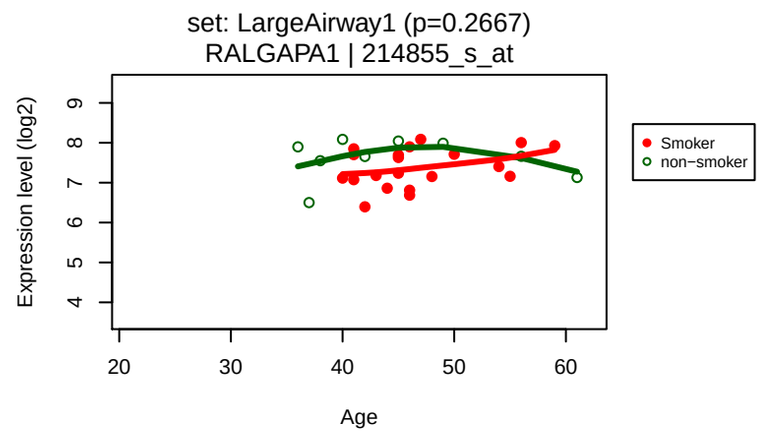

Gene expression of smokers and nonsmokers
gene: RALGAPA1 | 214855_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2127
p=0.3052
p=0.0492
p=0.2667
p=0.0944
p=0.653
p=0.8287
p=0.8756
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

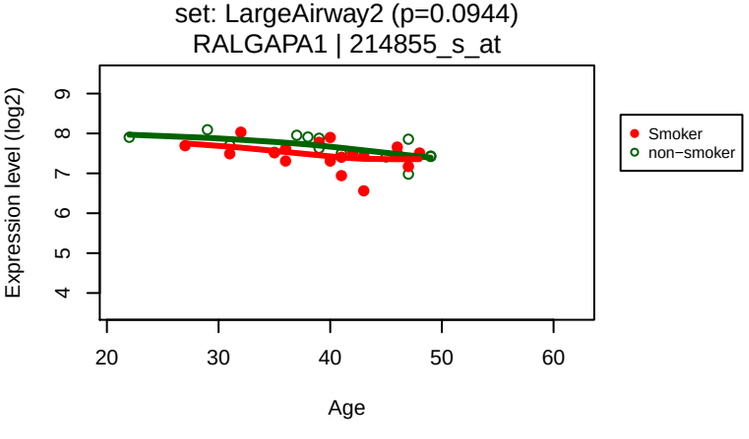

Gene expression ratio (smokers/nonsmokers)
gene: RALGAPA1 | 214855_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2127
p=0.3052
p=0.0492
p=0.2667
p=0.0944
p=0.653
p=0.8287
p=0.8756
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

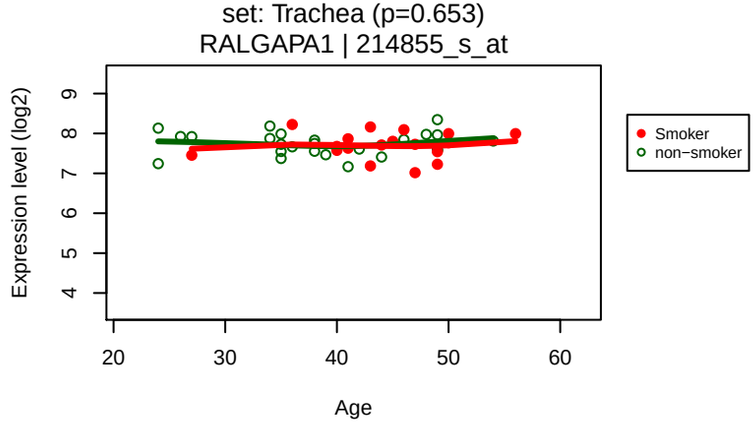

Gene expression ratio in Males
gene: RALGAPA1 | 214855_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4967
p=0.1619
p=0.7248
p=0.8676
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALGAPA1 | 214855_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.699
p=0.7466
p=0.8246
p=0.6995
SmallAirway2
LargeAirway2
Trachea
Macrophage2

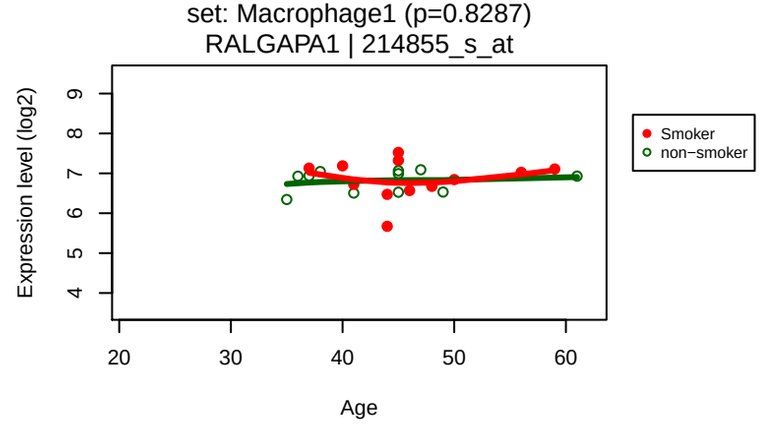

set: SmallAirway1 (p=0.2127)
RALGAPA1 | 214855_s_at
Expression level (log2)
Age
Smoker
non−smoker

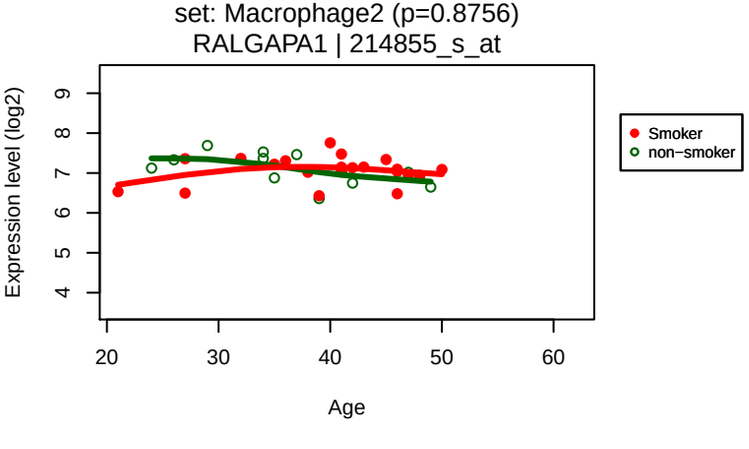

set: SmallAirway2 (p=0.3052)
RALGAPA1 | 214855_s_at
Expression level (log2)
Age
Smoker
non−smoker

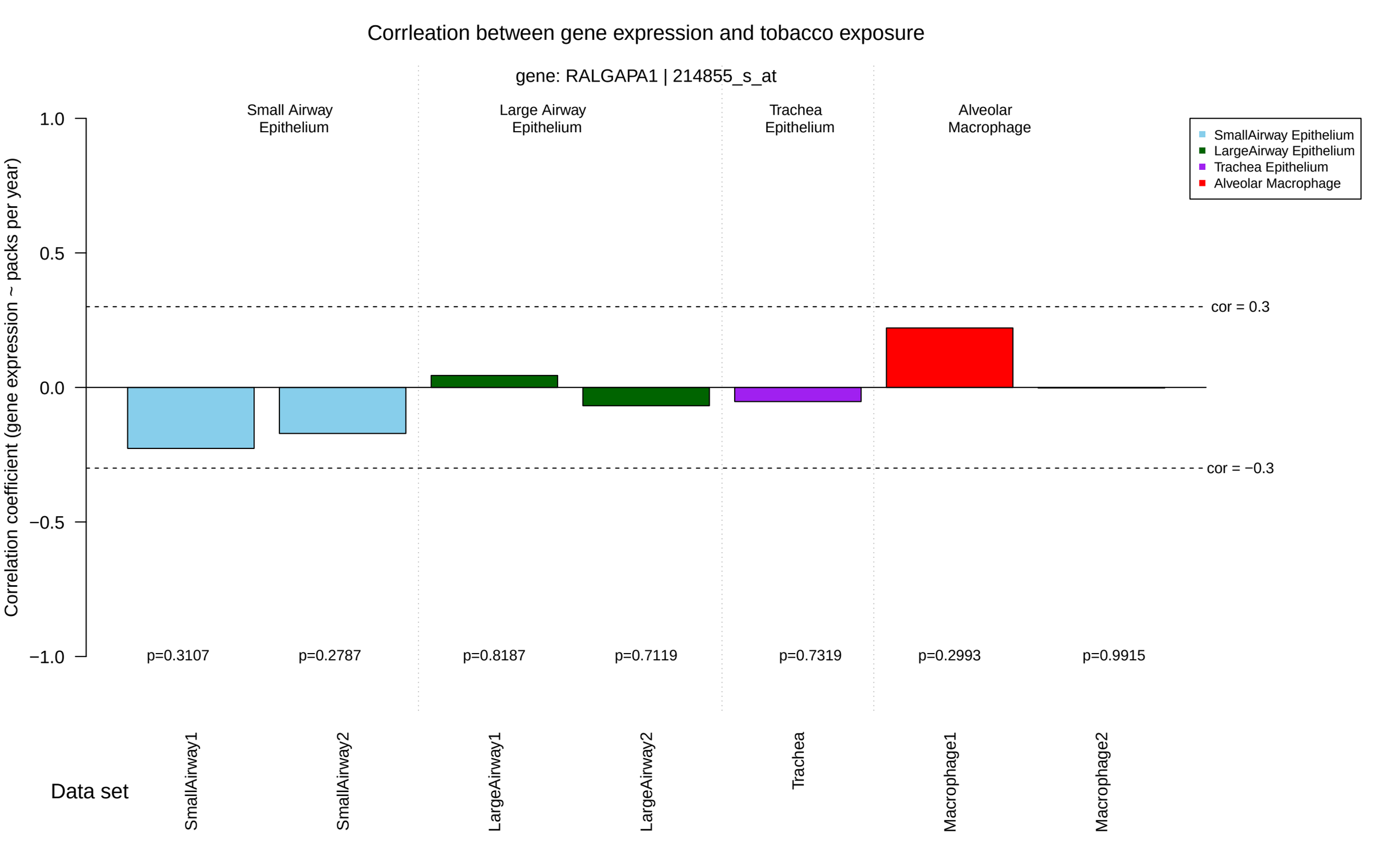

set: LargeAirway1 (p=0.2667)
RALGAPA1 | 214855_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0944)
RALGAPA1 | 214855_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.653)
RALGAPA1 | 214855_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8287)
RALGAPA1 | 214855_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8756)
RALGAPA1 | 214855_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALGAPA1 | 214855_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3107
p=0.2787
p=0.8187
p=0.7119
p=0.7319
p=0.2993
p=0.9915
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2